Fiscal Year:	FY 2020	**Task Last Updated:**	FY 06/11/2020
PI Name:	McLean, Robert Ph.D.		
Project Title:	Polymicrobial Biofilm Growth and Control during Spaceflight		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78666-4684	**Congressional District:**	35
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	12/05/2016	**End Date:**	12/04/2020
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	December 2019 report: Key personnel on the original grant application are unchanged. This includes co-Principal Investigator (PI) Cheryl Nickerson and her team (Jennifer Barrila and Jiseon Yang) at Arizona State University, as well as C Mark Ott (NASA-Johnson Space Center-JSC). Two additional participants, Sandhya Gangaraju and Richard Davis at Arizona State University are also participating. Due to the expertise of our team, we are able to do all the necessary microscopy and will not require external collaborators as originally considered. Starla Thornhill was a masters' student as the grant was being written. She successfully defended her MS thesis in 2016 and is now doing her PhD on work directly related to this grant. I had one MS student, Quentin DiPasquale (now graduated), who did modeled microgravity work that is peripheral to the NASA-funded project. With Quentin's graduation, I have another new MS student, Calvin Tran, who is continuing the microgravity work to see if aquatic organisms preferentially grow in simulated microgravity. As well, we have added Alistair McLean as an unpaid collaborator. He works as a software engineer and is writing a program that will help us enumerate bacterial populations from microscopy images.		
COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, C. Mark Ph.D. (NASA Johnson Space Center) Nickerson, Cheryl Ph.D. (Arizona State University)		
Grant/Contract No.:	NNX17AC79G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Bacteria and other microorganisms have been shown to grow as surface-adherent biofilm communities in the microgravity environment during spaceflight. Organisms within biofilms are quite resistant to traditional antimicrobial chemicals and can foul water treatment filters resulting in potential risk to the astronauts. The biofilms on International Space Station (ISS) consist of many species, yet studies to date have only tested single-species biofilms. There is growing evidence that mixed-culture biofilms behave quite differently from pure culture biofilms, and we anticipate this phenomenon to occur in the microgravity environment of ISS. Through this study we will investigate the structure by which two-species, mixed-culture biofilms form under microgravity on stainless steel used to construct Environmental Controls and Life Support Systems (ECLSS) on the ISS. As well, we will explore the efficacy by which mixed-culture biofilms can be controlled, and whether microgravity-grown mixed-culture biofilms cause corrosion on stainless steel. As a result of supplemental funding, we are now able to gather genetic data on the mechanisms by which mixed bacterial species are able to form biofilms in microgravity conditions. Our results are designed to both enable future spaceflight missions and translate into information for the general scientific community on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In work accomplished to date, we have successfully shown that mixtures of different bacteria can grow and form aggregates on surfaces and in liquid (bacterial biofilms), when grown under conditions that mimic the water recycling equipment on the ISS. Our experiments involved using the bacteria Escherichia coli (E. coli) and Pseudomonas aeruginosa (P. aeruginosa). As well, we have just inserted genes that code for fluorescent proteins into each of these organisms so that we can identify and distinguish them using a microscope as E. coli will fluoresce red and P. aeruginosa will fluoresce green. We are also developing a software program that can be used to count and analyze differently stained bacteria in microscopy images. This program is intended to be freely available to the scientific community. These two organisms, E. coli and P. aeruginosa, are often found in wastewater recycling systems and our future work will let us determine whether they need to grow together in a microgravity environment, and does one organism protect another if a disinfectant is used. That protection appears to occur in several microbial fouling environments on Earth and one benefit of the proposed work will be to investigate how reduced gravity affects bacterial protection against a proposed silver-based disinfectant. Through additional funding, we are now able to do genetic analysis of the two bacteria under microgravity conditions and so gain insight as to the mechanisms by which they are able to form biofilms and the effect of the disinfectant on these mechanisms, in comparison to full-gravity control experiments done on Earth. We have also completed some tests and observed that biofilm growth by the E. coli and P. aeruginosa microorganisms can cause some slight corrosion of stainless steel. In planned experiments during the coming year, we will investigate if gravity or reduced gravity has an influence on this process.
Task Progress:	In the first year of the grant, we purchased necessary equipment for conducting simulated weightlessness (modeled microgravity) experiments in the lab. We tested a number of bacterial strains to identify two representative bacteria, Escherichia coli (E. coli) and Pseudomonas aeruginosa (P. aeruginosa). While the P. aeruginosa strain PAO-1 had been grown successfully in several ground-based modeled microgravity and flight-based studies, the E. coli strains (originally associated with urinary tract infections) had never been studied, although there is at least one incident (Apollo 13) in which an astronaut had a urinary infection, so the choice of an organism with that infection potential is quite relevant. From 12 candidate E. coli strains, we selected one, E. coli strain F11, which had originally been isolated from a patient with a first episode of a bladder infection. The criteria for selection was the ability of these two strains to grow in a modified artificial urine medium (mAUM) under modeled microgravity (MMG) at ambient temperature (25C), which mimics conditions on the ISS. Based on the ability of these bacteria to grow as biofilms in mAUM in modeled microgravity, we selected our bacterial strains (P. aeruginosa PA01 and E. coli F11). The P. aeruginosa strain contained the gene for the green fluorescent protein (gfp) and we inserted a gene for a red-fluorescing protein (mCherry) into the E. coli F11 strain so that these two bacteria could be distinguished by microscopy as green- and red-fluorescing cells. During year two, we progressed towards getting flight readiness. Much of our experimental work has been done in concert with Bioserve at the University of Colorado, who are doing the hardware (Biocell) preparation that will be used for flight experiments. Variations in flight hardware were tested with the key issues being biological reproducibility and the ability of the specimens to be analyzed by confocal microscopy. Confocal microscopy is designed to acquire a three-dimensional image of biological structures and the individual bacteria could be identified on the basis of the green- and red-fluorescing proteins. We also obtained different stains that could identify surface materials (slime) produced by the two bacteria. When we first started the project, we had planned on using microscopy facilities available on the ISS. However, we learned that the ISS confocal microscope would not be able to illuminate one of the key fluorescent proteins (gfp) and so had to adjust our planning and experimentation to preserve the bacteria upon completion of the experiment during flight, and then doing all microscopy after the completion of the flight. During year three (current report), we finalized the flight hardware design, and identified the three time points for data collection. Initial difficulties with the stainless steel component of the flight hardware were fixed with the help of NASA Marshall Space Flight Center (NASA-MSFC). We were fortunate to get some modest additional support that now enables us to do genetic analysis of our samples and identify mechanisms by which bacteria are able to form mixed-species biofilms in microgravity in the presence and absence of a disinfectant. We were able to finalize the experimental conditions and identify some restrictions (such as stability of the media and microorganisms prior to experimentation, and the stability of the fluorescent signal and genetic material (RNA) following experimentation)). One notable success is our microscopy. We are now able to do all the necessary microscopy (both confocal microscopy and electron microscopy). At the conclusion of year three, we were in a position to do an overall evaluation of the entire procedure (called an Experimental Verification Test (EVT)) using materials and equipment intended for use by the space crew. Ongoing work for year four: EVT planning and experimentation (now completed); Establishing a timeline for experimentation (now completed); Training of additional team members for microscopy (in progress); Flight experiment (our experiment has been delayed

	until likely Space X-21) and planning is in progress for that as well as post-flight microscopy and genetic analysis, and overall data analysis.
Bibliography Type:	Description: (Last Updated: 11/20/2025)
Articles in Peer-reviewed Journals	Zea L, McLean RJC, Rook TA, Angle G, Carter DL, Delegard A, Denvir A, Gerlach R, Gorti S, McIlwaine D, Nur M, Peyton BM, Stewart PS, Sturman P, Velez Justiniano YA. "Potential biofilm control strategies for extended spaceflight missions." Biofilm. 2020 Dec;2:100026. First Available online 30 May 2020. PMID: 33447811; PMCID: PMC7798464 ; https://doi.org/10.1016/j.bioflm.2020.100026 , Dec-2020